COVERS AND DEGREE BOUNDS FOR A QUASI-COMPACT QUASI-SEPARATED SPACE, AND THE RESULTING VANISHING ABOVE THE MAXIMUM OF THE BOUNDS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- An affine etale cover, and the same applied to find affine covers of the configuration quotients. Properties of Spaces, label spaces-properties-lemma-quasi-compact-affine-cover (invoked at proof lines 1406 and 1419). It supplies the affine U mapping surjectively and etale onto X, and later the affine schemes V_p mapping onto the configuration quotients U_p.
- Universal boundedness of the fibres of a morphism from a scheme to a reasonable algebraic space. Decent Spaces, label decent-spaces-lemma-bounded-fibres (invoked at proof lines 1409 and 1422). It produces the integer d bounding the degrees of the fibres of $U \to X$ and the integers d_p bounding the degrees of $V_p \to U_p$.
- Quasi-compactness and separatedness of the quotients of the configuration schemes. Spaces, label spaces-lemma-quotient-finite-separated (invoked at proof line 1416). Since X is quasi-separated the schemes W_p are quasi-compact and separated, hence the quotients $U_p = W_p/S_{p+1}$ are too.
- The alternating Cech spectral sequence, the vanishing on the separated case, and quasi-coherence of the twists. Same chapter spaces-cohomology.tex, labels lemma-alternating-spectral-sequence (invoked at proof line 1434), proposition-vanishing (invoked at proof line 1436) and lemma-quasi-coherent-twist (invoked at proof line 1438). The spectral sequence computes the cohomology of X from the twisted cohomology of the U_p; the second ingredient makes those groups vanish above d_p; the third keeps the twisted sheaves quasi-coherent so that the vanishing applies.

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Source reference. Original lemma 03JX: sources/stacks-chow/tags.txt, source lines 4655–4655. Navigation pointer for \ref{decent-spaces-lemma-bounded-fibres} used inside the counted span (line 1409); the argument that consumes it is retained in the span.

Source reference. Original lemma 0728: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1217–1217. Navigation pointer for \ref{lemma-alternating-spectral-sequence} used inside the counted span (line 1395); the argument that consumes it is retained in the span.

Source reference. Original lemma 072A: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1305–1305. Navigation pointer for \ref{lemma-quasi-coherent-twist} used inside the counted span (line 1438); the argument that consumes it is retained in the span.

Source reference. Original proposition 072B: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1333–1333. Navigation pointer for \ref{proposition-vanishing} used inside the counted span (line 1436); the argument that consumes it is retained in the span.

Source reference. Original lemma 02Z4: sources/stacks-chow/tags.txt, source lines 3899–3899. Navigation pointer for \ref{spaces-lemma-quotient-finite-separated} used inside the counted span (line 1416); the argument that consumes it is retained in the span.

Source reference. Original lemma 03H6: sources/stacks-chow/tags.txt, source lines 4555–4555. Navigation pointer for \ref{spaces-properties-lemma-quasi-compact-affine-cover} used inside the counted span (line 1407); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Assume X is quasi-compact and quasi-separated. Then we can choose*

1. *an affine scheme U,*
2. *a surjective étale morphism $f : U \to X$,*
3. *an integer d bounding the degrees of the fibres of $U \to X$,*
4. *for every $p = 0, 1, \ldots, d$ a surjective étale morphism $V_p \to U_p$ from an affine scheme V_p where U_p is as in Lemma 0728, and*
5. *an integer d_p bounding the degree of the fibres of $V_p \to U_p$.*

Moreover, whenever we have (1) – (5), then for any quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ we have $H^q(X, \mathcal{F}) = 0$ for $q \geq \max(d_p + p)$.

Proof. Since X is quasi-compact we can find a surjective étale morphism $U \to X$ with U affine, see Properties of Spaces, Lemma 03H6. By Decent Spaces, Lemma 03JX the fibres of f are universally bounded, hence we can find d. We have $U_p = W_p/S_{p+1}$ and $W_p \subset U \times_X \ldots \times_X U$ is open and closed. Since X is quasi-separated the schemes W_p are quasi-compact, hence U_p is quasi-compact. Since U is separated, the schemes W_p are separated, hence U_p is separated by (the absolute version of) Spaces, Lemma 02Z4. By Properties of Spaces, Lemma 03H6 we can find the morphisms $V_p \to W_p$. By Decent Spaces, Lemma 03JX we can find the integers d_p.

At this point the proof uses the spectral sequence

$$E_1^{p,q} = H^q(U_p, \mathcal{F}|_{U_p} \otimes_{\mathbf{Z}} \underline{\mathbf{Z}}(\chi_p)) \Rightarrow H^{p+q}(X, \mathcal{F})$$

see Lemma 0728. By definition of the integer d we see that $U_p = 0$ for $p \geq d$. By Proposition 072B and Lemma 072A we see that $H^q(U_p, \mathcal{F}|_{U_p} \otimes_{\mathbf{Z}} \underline{\mathbf{Z}}(\chi_p))$ is zero for $q \geq d_p$ for $p = 0, \ldots, d$. Whence the lemma. □